Mathematical methods for continuum modelling
Matrices

Stephen Millmore

Laboratory for Scientific Computing, University of Cambridge

Outline

1. Properties of matrices
2. Elementary operations
3. The inverse
4. The determinant
5. Eigenvalues and eigenvectors
6. LU decomposition

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Matrices

- We briefly mentioned matrices when talking about index notation. Now we shall go into more details about their properties
- The key focus in this topic is matrices used for solving systems of equations, where it is more common to work with larger $(N \times N)$ systems than the three-dimensional systems we considered under vector calculus
- The fact that such matrices are encountered throughout scientific computing is probably not a surprise
- They are used in solving linear systems of equations (e.g. incompressible flow) or analysing the properties of systems of partial differential equations
- We could argue that the only things we need to know about matrices are how to find eigenvalues (and eigenvectors) and how to compute the inverse
- In scientific computing, though, we may wish to do these things quickly, and that is not always simple

Matrices and vectors

- When understanding properties of matrices, or when converting a system of equations into matrix form, it can be convenient to think of matrices as a collection of vectors
- For example,

$$A = (\mathbf{v}_1 | \mathbf{v}_2 | \mathbf{v}_3 | \cdots | \mathbf{v}_M)$$

  where $\mathbf{v}_i$ is a column vector
- Or, if $\mathbf{w}_j$ is a row vector

$$B = \begin{pmatrix} \mathbf{w_1} \\ \hline \mathbf{w_2} \\ \hline \mathbf{w_3} \\ \hline \vdots \\ \hline \mathbf{w_M} \end{pmatrix}$$

Linear dependency

- Many key matrix properties are related to the idea of linear dependency in the matrix rows or columns
- Rows are linearly dependent if one row of the matrix can be written as a linear combination of other rows
- Mathematically, we say that a matrix is linearly independent if

$$a_1\mathbf{v}_1 + a_2\mathbf{v_2} + a_3\mathbf{v}_3 + \cdots + a_M\mathbf{v_M} = \mathbf{0}$$

  is satisfied only if $a_i = 0$ for all i
- Otherwise, if non-zero values exist which satisfy this relationship, we could write

$$\mathbf{v}_1 = \frac{1}{a_1}\left(a_2\mathbf{v_2} + a_3\mathbf{v}_3 + \cdots + a_M\mathbf{v_M}\right) \tag{1.1}$$

- That is, we can write one row as a combination of the other rows (or, equivalently, columns)

Familiar types of square matrix

- Symmetric matrices: $A_{ij} = A_{ji}$
- Antisymmetric matrices: $A_{ij} = -A_{ji}$
- Diagonal matrices: $A_{ij} = 0$ if $i \neq j$
- Upper triangular matrices:

$$B_{ij} = \begin{cases} b_{ij} & i \leq j \\ 0 & i > j \end{cases}$$

- Lower triangular matrices:

$$C_{ij} = \begin{cases} c_{ij} & i \geq j \\ 0 & i < j \end{cases}$$

Familiar square matrix properties

- Trace of a matrix: $\mathrm{Tr}(\mathrm{A}) = A_{ii}$
- Inverse, A^{-1}: $(\mathrm{A}^{-1})_{ik} A_{kj} = \delta_{ij}$‘
- Determinant:

$$\det(A) = \sum_k (-1)^{1+k} M_{1,k} A_{1k}$$

  where $M_{i,j}$ is **minor** of A, which is the determinant of the matrix with the ith row and j^{th} column removed (there is no neat, single-line expression to use)

- It is hard (but not impossible), and usually impractical, to write the determinant in index notation form

$$\det(A) = \epsilon_{i_1 i_2 i_3 \ldots i_N} A_{1 i_1} A_{2 i_2} A_{3 i_3} \ldots A_{N i_N}$$

  for an $N \times N$ matrix

- An extension of the Levi-Civita symbol has been used here

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Elementary row and column operations

- When working with matrices, we often need to move rows and/or columns about
- This might be when solving simultaneous equations, computing a determinant or inverting a matrix
- These are likely very familiar in the first application

$$\begin{aligned} ax + by + cz &= j \\ dx + ey + fz &= k \\ gx + hy + iz &= l \end{aligned} \qquad \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} j \\ k \\ l \end{pmatrix}$$

- When solving this system, it is likely you will use three operations:
  1. Multiplying a row by a scalar: $r_i \to \lambda r_i$
  2. Adding a multiple of one row to another: $r_i \to r_i + \mu r_j$
  3. Exchanging rows: $r_i \leftrightarrow r_j$
- These are **elementary row operations** or simply **row operations**

Elementary row (column) operation matrices

- Each of the elementary row (or column) operations can be considered as a matrix which multiplies our initial matrix, or equation, from the left (or right, for column operations)
- $r_2 \to \lambda r_2$:

$$\mathrm{E}_\lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- $r_1 \to r_1 + \mu r_3$:

$$\mathrm{E}_\mu = \begin{pmatrix} 1 & 0 & \mu \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- $r_2 \leftrightarrow r_3$:

$$\mathrm{E}_\nu = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Elementary matrix notation

- Typically, multiple elementary operations are done on a single matrix, each operation denoted E_α
- This can be denoted

$$\mathrm{A}' = \mathrm{E}_n \mathrm{E}_{n-1} \ldots \mathrm{E}_2 \mathrm{E}_1 \mathrm{A} = \mathrm{E}\mathrm{A}$$

- Note that E, the matrix formed by multiplying the elementary operation matrices, is not necessarily a unique combination of these operations

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Properties of the inverse

- Commutativity: $\mathrm{A}\mathrm{A}^{-1} = I$ and $\mathrm{A}^{-1}\mathrm{A} = I$
- Inverse of an inverse: $(\mathrm{A}^{-1})^{-1} = \mathrm{A}$
- Scalar multiplication: $(k\mathrm{A})^{-1} = \frac{1}{k}\mathrm{A}^{-1}$
- Inverse of a product: $(\mathrm{A}\mathrm{B})^{-1} = \mathrm{B}^{-1}\mathrm{A}^{-1}$
- Which extends to:

$$(\mathrm{A}_1\mathrm{A}_2 \dots \mathrm{A}_n)^{-1} = (\mathrm{A}_n)^{-1} \dots (\mathrm{A}_2)^{-1}(\mathrm{A}_1)^{-1}$$

- Not every matrix has an inverse, but the elementary matrices do

Inverse of the elementary matrices

- In order to show further properties of the inverse, and matrices in general, it is important to show that elementary matrices are invertible
- This is probably not too surprising - it is equivalent to row (or column) operations being reversible, i.e. we have E_1 and E_2 such that

$$\mathrm{E}_2\mathrm{E}_1\mathrm{A} = \mathrm{A}$$

- Multiplication of a row by a scalar is reversed by division of a row by the same scalar
- Adding a multiple of a row is reversed by subtracting the same multiple of the row
- Switching two rows is reversed by switching them again
- Because all these operations are possible, we must have

$$\mathrm{E}_2 = (\mathrm{E}_1)^{-1}$$

Non-invertible matrices

- If we consider the simultaneous equation $A\mathbf{x} = \mathbf{b}$, then we can write

$$\mathrm{A}^{-1}\mathrm{A}\mathbf{x} = \mathbf{x} = \mathrm{A}^{-1}\mathbf{b}$$

- This implies that there is a unique solution for $\mathbf{x}$
- We know that this is not true for all simultaneous equations
- In this case, A^{-1} does not exist
- Alternatively, a system of simultaneous equations does not give a unique solution when one of the equations can be written as a combination of the others - when there is linear dependency
- We can show that linearly dependent matrices do not have an inverse

Invertible matrices

- The converse to the previous slide is also true - linearly independent matrices have an inverse
- We can show that for a linearly independent matrix, a series of row operations exist such that

$$\mathrm{EA} = \mathrm{I} \tag{3.1}$$

- This is done by first putting our matrix A in row echelon form, and when we do this, we know that all diagonal entries must be non-zero
- In doing so, we find that $\mathrm{A}^{-1} = \mathrm{E}$
- This then also gives us a technique to find the inverse

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Determinants of diagonal matrices

- The determinant of a diagonal matrix, or an upper or lower triangular matrix, is straightforward to compute - simply multiply the diagonal entries together
- When computing the determinant of other matrices, especially those with algebraic entries, rather than just numbers, this can be useful
- We know we can convert a matrix to row echelon form, and from that, the determinant is now easy to compute
- All we need to know is what effect the elementary operations we used to obtain row echelon form have on the determinant
- Fortunately, this is easy to work out

Effect of elementary operations on determinants

- Multiplying a row by a scalar multiplies the determinant by the same scalar

$$\mathrm{A}' = \mathrm{E}_\lambda \mathrm{A} \quad \Longrightarrow \quad \det(\mathrm{A}') = \lambda \det(\mathrm{A})$$

- Adding a multiple of one row to another doesn't change the determinant

$$\mathrm{B}' = \mathrm{E}_\mu \mathrm{B} \quad \Longrightarrow \quad \det(\mathrm{B}') = \det(\mathrm{B})$$

- Switching two rows switches the sign of the determinant

$$\mathrm{C}' = \mathrm{E}_\nu A \quad \Longrightarrow \quad \det(\mathrm{C}') = -\det(\mathrm{C})$$

- This means a multiple elementary operations either multiply the determinant by a scalar (including -1) or leave it unchanged
- For an arbitrary number of operations, E, we have

$$\mathrm{A}' = \mathrm{E}\mathrm{A} \quad \Longrightarrow \quad \det(\mathrm{A}') = \alpha \det(\mathrm{A})$$

Determinants and linear dependency

- If a matrix has a row which is entirely zeroes, it is easy to see (and show) that the determinant is zero
- For a linearly dependent matrix, we can perform elementary operations,

$$\mathrm{A}' = \mathrm{EA}$$

  such that A' contains a row of zeroes
- Therefore

$$\det(\mathrm{A}') = 0 \qquad \text{and} \qquad \det(\mathrm{A}') = \alpha \det(\mathrm{A})$$

- This means $\det(\mathrm{A}) = 0$ if A has linearly dependent rows or columns
- The converse also holds - a linearly independent matrix, B, has $\det(\mathrm{B}) \neq 0$
- As a result, the determinant is a quick method to check for linear dependency

Determinant of a product

- The determinant of a matrix product obeys

$$\det(\mathrm{AB}) = \det(\mathrm{A})\det(\mathrm{B})$$

- Although a simple expression, it is not obvious to prove
- We consider the cases of A being linearly dependent and independent separately
- **Linear dependency**: Since $\det(\mathrm{AB}) \neq 0$ would imply that $(\mathrm{AB})^{-1}$ exists, we should have

$$(\mathrm{AB})(\mathrm{AB})^{-1} = \mathrm{ABB}^{-1}A^{-1} = \mathrm{I}$$

  but since A^{-1} doesn't exist, this is a contradiction
- **Linear independence**: In this case, we know we have $\mathrm{A} = \mathrm{EI}$, so

$$\det(\mathrm{AB}) = \det(\mathrm{EB}) = \alpha\det(\mathrm{B})$$

- By choosing the case of $\mathrm{B} = \mathrm{I}$, we see $\alpha = \det(\mathrm{A})$

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Matrices as coordinate transformations

- A related concept to eigenvalues and eigenvectors is the fact that a vector multiplied by a matrix is a linear map of that vector to a new coordinate system
- We often don't need to consider this explicitly, but sometimes it is useful
- For example, if we have $\mathbf{v}' = \mathrm{E}_\nu \mathbf{v}$ with

$$\mathrm{E}_\nu = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

  we change the handedness of the coordinate system

- And for $\mathbf{v}' = \mathrm{R}\mathbf{v}$ with

$$\mathrm{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

  we rotate our coordinate system (around the z-axis)

Eigenvalues and eigenvectors

- Sometimes, multiplication of a vector by a matrix will not change the vector, or will simply scale it

$$\mathrm{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

- This can be very useful for making absolute statements about a system
- And this is behind the definition of eigenvalues and eigenvectors

$$A\mathbf{v} = \lambda\mathbf{v}$$

- Or we could have

$$\mathbf{w}^T A = \lambda\mathbf{w}^T$$

  which are the **left eigenvectors** of the system
- $\mathbf{v}$ in the expression above is a **right eigenvector**, or often, simply an eigenvector

Finding eigenvalues

$$\mathrm{A}\mathbf{v} = \lambda\mathbf{v}$$

- We want to find non-zero solutions to

$$(\mathrm{A} - \lambda\mathrm{I})\mathbf{v} = 0$$

- The zero vector, $\mathbf{0}$, is always trivially a solution - this is not an eigenvector
- The only way we can obtain non-zero solutions is if $(\mathrm{A} - \lambda\mathrm{I})$ is a linearly dependent matrix
- This gives us an easy way to find the eigenvalues

$$\det(\mathrm{A} - \lambda\mathrm{I}) = 0$$

- We get a polynomial in λ which is satisfied by the eigenvalues - this also tells us

$$\det(\mathrm{A}) = \lambda_1\lambda_2\lambda_3\cdots\lambda_N$$

Finding eigenvectors

- Once you have your eigenvalues, finding the eigenvectors is a matter of solving a simultaneous equation

$$\mathrm{A} = \begin{pmatrix} 0 & -4 & -3 \\ -3 & 1 & 3 \\ -2 & 1 & 4 \end{pmatrix}$$

- This matrix has eigenvalues $\lambda_1 = 1, \lambda_2 = -3, \lambda_3 = 7$
- Therefore one eigenvector is obtained by solving $(\mathrm{A} - 7\mathrm{I})\mathbf{v} = 0$,

$$\mathrm{A} = \begin{pmatrix} -7 & -4 & -3 \\ -3 & -6 & 3 \\ -2 & 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}$$

- We know that there is not a unique solution to this equation, and what we find is

$$\mathbf{v}_3 = a \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

for any constant a - it is common to choose a such that $\mathbf{v}$ is normalised

Finding eigenvalues - other possibilities

- Our first example considered three unique real eigenvalues
- There is no need for them to be either of these - complex eigenvalues are common, and result in complex eigenvectors, and repeated eigenvalues are common too (see next slide)
- The eigenvalues of diagonal matrices are particularly easy - they are the diagonal entries
- Elementary row operations do not change the eigenvalues of a matrix
- This means you can sometimes manipulate a matrix to make finding the eigenvalues easier
- Linearly dependent matrices always have at least one eigenvalue of zero

Finding eigenvectors - repeated eigenvalues

- We might find we have as many eigenvectors as we have unique eigenvalues

$$\mathrm{A} = \begin{pmatrix} 5 & -2 & 0 \\ 0 & 4 & 0 \\ 3 & -5 & 5 \end{pmatrix}$$

- This matrix has eigenvalues $\lambda_1 = 4, \lambda_{2,3} = 5$
- Here we find two eigenvalues,

$$\lambda_1 : \quad \hat{\mathbf{v}}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \qquad \lambda_2 : \quad \hat{\mathbf{v}}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Finding eigenvectors - repeated eigenvalues

- We might find we have as more eigenvectors than we have unique eigenvalues

$$\mathrm{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

- This matrix has eigenvalues $\lambda_1 = 2, \lambda_{2,3} = -1$
- Here we could say that we find two eigenvalues,

$$\lambda_1 : \quad \hat{\mathbf{v}}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \qquad \lambda_2 : \quad \hat{\mathbf{v}}_2 = \begin{pmatrix} x \\ y \\ -x - y \end{pmatrix}$$

- However, we could split $\hat{\mathbf{v}}_2$ into two vectors (one using x, one using y):

$$\hat{\mathbf{v}}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \hat{\mathbf{v}}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

Linear independence of eigenvectors

- Whilst it is clear that all the eigenvectors we have found so far are different, this doesn't necessarily mean they are linearly independent
- However, it is possible to show that they are - the eigenvectors from distinct eigenvalues are always linearly independent
- For cases where repeated eigenvalues can be written as more than one eigenvector, we can also choose these to be linearly independent
- A result of this is that two eigenvalues can never have the same eigenvector
- In fields where eigenvectors form a useful coordinate system, knowing that they are independent is particularly important

Diagonalisation

- Once we have identified the eigenvalues of a matrix, we know that a diagonal matrix, with these eigenvalues as the entries, would clearly have the same eigenvalues
- This means there is a technique, **diagonalisation** which allows you to connect these two matrices
- Specifically, we have

$$\mathrm{A} = \mathrm{S}\Lambda\mathrm{S}^{-1}$$

  where S is the matrix whose columns are the eigenvectors of A
- Not all matrices are diagonalisable, such as an $N \times N$ matrix which doesn't have N eigenvectors
- But where a matrix is diagonalisable, there are many useful implications, including being able to compute matrix powers

$$\mathrm{A}^m = \mathrm{S}\Lambda^m\mathrm{S}^{-1}$$

Similar matrices

- The relationship we saw for diagonalisable matrices does generalise - any matrices A and C related by

$$\mathrm{C} = \mathrm{BAB}^{-1}$$

  are **similar matrices**

- Similar matrices have some useful properties:
  1. They have the same determinant
  2. They have the same eigenvalues
  3. They have related eigenvectors (but not the same eigenvectors)
- When dealing with algebraic matrices in particular, similar matrices can be useful to make it easier to compute some of these properties

Symmetric matrices

- It is quite common for physical systems with matrix quantities to have symmetric matrices
- These have particularly useful properties which can greatly simplify computations
- The eigenvalues of a symmetric matrix are **always real** - this is very useful to know if you want to compute these computationally
- The eigenvectors of a symmetric matrix are always orthogonal - we can sometimes use this to create a coordinate system with useful physical properties

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Triangular matrices in simultaneous equations

- We know that we can use various matrix operations to solve simulatneous equations

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

- However, in scientific computing, A could be large - algorithms which work through row reduction operations can be very expensive
- Sometimes A has nice properties, however

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

- Here, we can read off the solution for z, and use this directly to find y, and then use that to find x (and similar for higher dimensions)
- Lower triangular matrices have similar nice properties

LU decomposition

- For any matrix, we can perform elementary operations to obtain

$$\mathrm{A} = \mathrm{EU}$$

  where U is an upper-triangular matrix
- Now, **if** none of the elementary operations were row-switching operations, then we have

$$\mathrm{E} = \mathrm{L}$$

  a lower-triangular matrix
- This is the LU decomposition, we can write

$$\mathrm{A} = \mathrm{LU}$$

- In this case, solving the simultaneous equations is now striaghtforward

Using LU decomposition

- From $\mathrm{A}\mathbf{x} = \mathbf{b}$ we have
$$\mathrm{LU}\mathbf{x} = \mathbf{b}$$
- Now, set $\mathbf{y} = U\mathbf{x}$,
$$\mathrm{L}\mathbf{y} = \mathbf{b}$$
- This is a lower-triangular simultaneous equation, it is easy to solve for $\mathbf{y}$
- But now, $\mathbf{y} = \mathrm{U}\mathbf{x}$ is an upper-triangular simultaneous equation with a known $\mathbf{y}$ - this is also easy to solve
- So, provided L and U were constructed with the right properties, instead of row reductions on A to solve the equation, we did row reductions on A to get U (and L at the same time), and then used this to solve the equation
- For a single equation, this hasn't gained us a lot, but...

Applications of LU decomposition

- It is common to need to solve $\mathrm{A}\mathbf{x} = \mathbf{b}$ repeatedly for different values of $\mathbf{x}$ but a single A
- In this case, LU decomposition means we only have to perform the complex elemetary operations once, rather than for every equation
- This can save us a lot of time in solving these equations computationally
- However, the chance that our elementary operations don't involve row switching are quite small
- In this case, though, we first work out what switches are required to obtain U, and then perform these on the system before the LU decomposition, and which point we can compute L and U for this newly ordered system